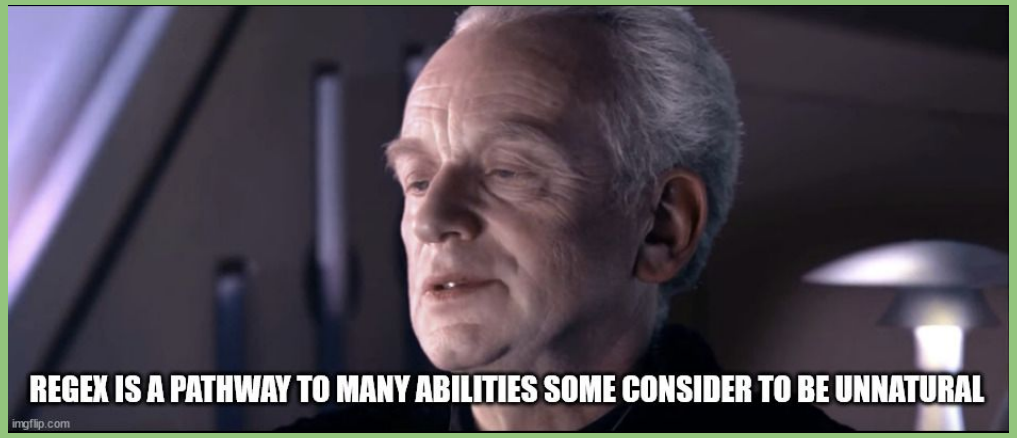
COMP2041 Week 2



Thank you Kyu-Sang Kim for the foundation of these slides!

Announcements

- **Welcome to Week 2 of COMP2041**

- Help Sessions start some time in late Week 3 or early Week 4
  - Schedule probably released in Week 3

- Week 1 & 2 labs due Week 3 Monday 12PM (midday) AEST
  - Lab due dates expected to normalise from Week 3

Overview

- Delimiters
- Useful UNIX Tools
- UNIX Pipelines “|”
- Tutorial Questions

Delimiters

- A delimiter is a specific character that separates fields in a line of text
- Common delimiters include: pipe (|), comma (,), colon (:), any amount of whitespace ( )
- Tools like cut and sort capitalise on this, allowing us to specify what fields to manipulate
- Example of delimited data: **Nick|21|75|COMP2041**
  - 4 fields (name, age, WAM, course) seperated by pipe (|).

Useful Unix Tools (WIP)

- sort
- head/tail
- cut
- tr
- grep
- sed
- uniq
- wc
- join
- man

Cut command

- Extracts specific fields from each line
- -d’**<delimiter>**’, sets the delimiter (default is **a tab**)
- -f’**<n>**’, selects which field(s) to extract
  - -f1 - extract the first field
  - -f1,f3 - extract the first and third field
- Example of use: **cut -d'|' -f1 data.txt**
  - Extracts specifically the first field, on data that is delimited by a pipe ‘|’

Sort command

- Sorts lines of text alphabetically by default
  - Can even sort based on a specific field in delimited data
- -t sets the delimiter (same idea as -d in cut)
- -k1**<N>** selects which field to sort by
  - Typically, you would do something like sort -k1, to sort field 1, however we should do **sort -k1,1.**
- -n sorts numerically instead of alphabetically
- -r sorts in reverse order
- Example of use: **cut -t'|' -k1 data.txt**
  - Sorts based on the first field, on data that is delimited by a pipe '|'

Head / Tail command

head

- Outputs the first N lines of a file
- Default is 10 lines
- **Head -n,** specifies how many lines to show

tail

- Exact same as head, but outputs the **last N** lines of file

tr command

- **Translates** or **deletes** characters
- Reads from stdin only (so must use I/O operators)

Character mapping rules

- **tr 'abcde' '123'**
  - **a→1, b→2, c→3, d→3, e→3**
- Replace group (second string) maps index to index with capture group (first string)
- If replace group is one character, everything in capture group is replaced by that character.

tr command (cont.)

- -d deletes specified characters
  - tr -d '|' < data.txt -> Deletes all pipes.
- -s squeezes repeated characters into one
  - tr -s ' ' < data.txt -> Turns all amount of continuous spaces, into one space
  - ‘ ‘ -> ‘ ‘
- -c complements the capture group — matches everything not in the set
  - tr -cd '0-9' < data.txt -> Delete everything that’s not a digit

sed command

- **Transform text line by line.**
  - In contrast to tr, it replaces strings with strings, rather than characters with characters.
- **Substitution command**
  - **sed -E** s/<pattern>/<replacement>/ — finds <pattern> and replaces with <replacement>
  - By default, only replaces first match per line. To ensure it replaces every match in file, do:
    - sed -E s/<pattern>/<replacement>/**g** -> Add the ‘global’ command at the end
- **Delete command**
  - **sed -E** /pattern/d — deletes every line that matches the pattern

sed command (cont.)

- By default, sed will always output the contents of the file after transformation.
- Similar to how we teach grep, we recommend using **-E** any time you use sed for access to extended regular expressions.
- **Print command**
  - /pattern/p — prints matching lines (usually paired with -n to suppress default output)
  - Without -n, matching lines print twice (because of the first dot point)

wc (Word Count)

- Word count — counts lines, words, and characters in a file
  - wc -l counts lines
  - wc -w counts words
  - wc -c counts characters (bytes)
- By default, wc will print **lines words characters** in that order

UNIX Pipelines

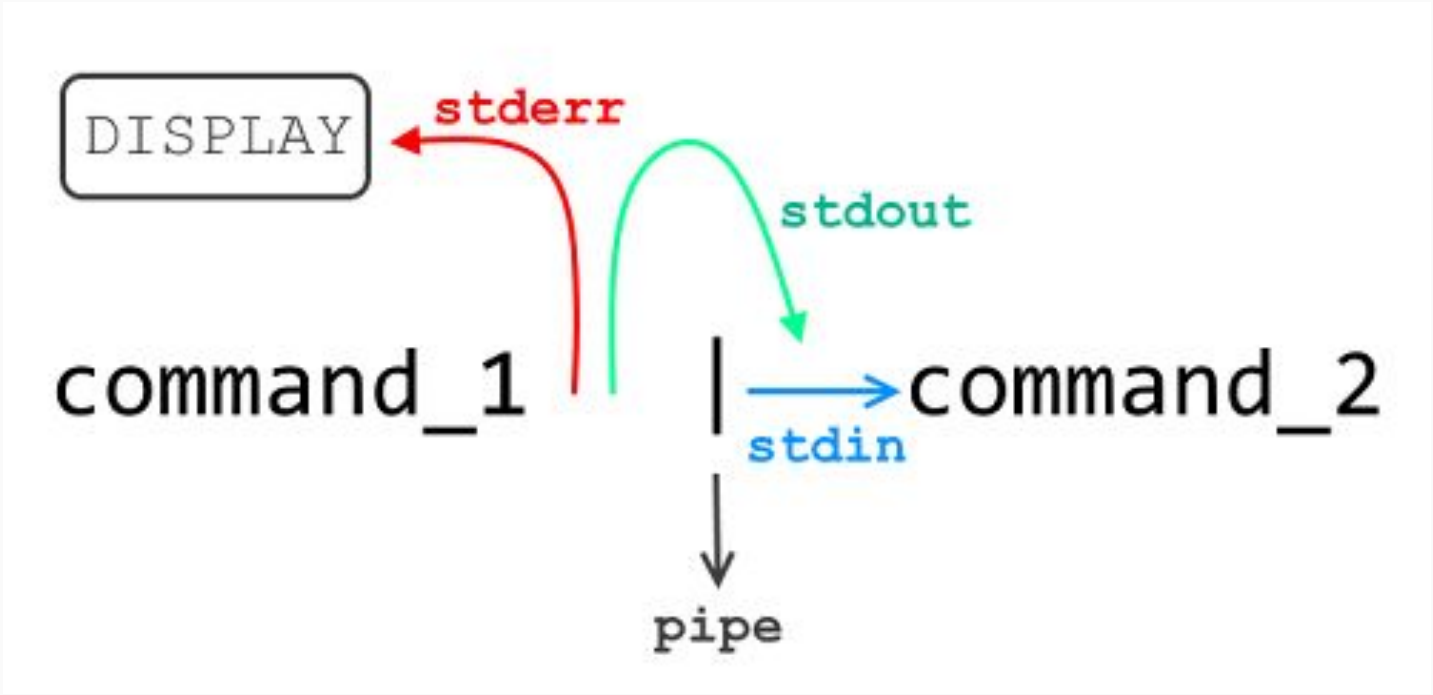
UNIX Pipelines

- Output of command passed to the next command
  - No need to save the output to a file

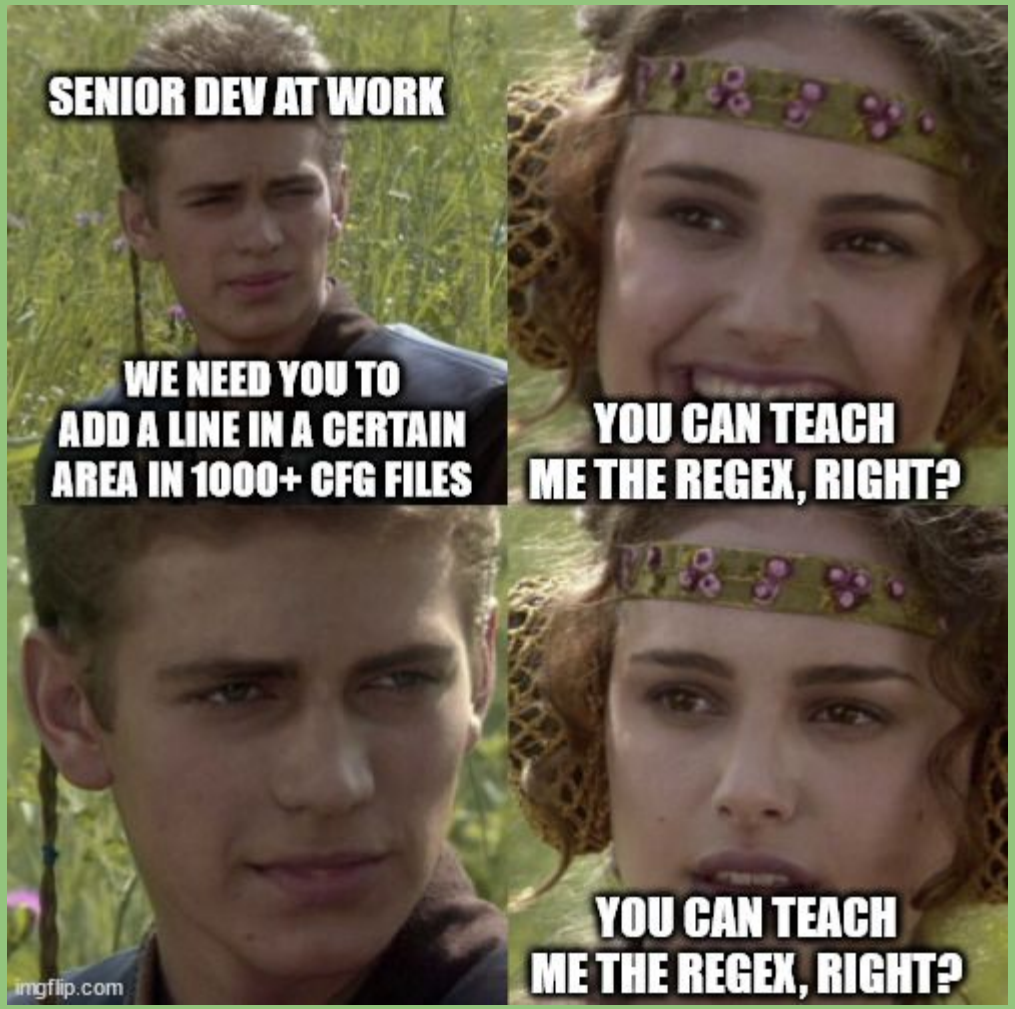
- Chained together to fulfil a much more complex process

- Data Parsers in their simplest form work in the same way
  - Improvements such as parallelism, clustering etc. are added on top

UNIX Pipelines



Tutorial Questions



To The Lab!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